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EXECUTIVE SUMMARY OF THE THESIS

Investigating Multimodal Contributions in Large Language Models Using audioDIME

LAUREA MAGISTRALE IN MUSIC AND ACOUSTIC ENGINEERING

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1. Introduction

Large Audio Language Models are increasingly able to answer questions about music, but correct answers alone do not demonstrate genuine listening. A model may solve a Music Question Answering (MQA) task by grounding its decision in the acoustic signal, or it may exploit textual priors, music-theoretical knowledge, answer-option regularities, and biases acquired during pretraining. This distinction is central in the evaluation of multimodal audio-language systems: the presence of an audio input does not necessarily imply that the model uses it in a causally relevant way.

This thesis investigates this problem in the context of Qwen2.5-Omni-7B [5], analysing whether the model actually relies on the audio signal when solving MQA tasks, or whether its decisions are mainly driven by the textual prompt. The work is motivated by the broader limitation of accuracy-based evaluation: a model can provide the correct answer while relying on shortcuts that are not perceptually grounded. For this reason, the thesis adopts an explainability-oriented perspective, aiming to analyse not only whether the model is right, but how its answer is produced.

The study focuses on HumMusQA [4], an expert-written MQA benchmark designed to reduce textual shortcuts and require perceptual musical reasoning. Its multiple-choice format provides a stable interpretative target, since the model's decision can be analysed through the logits of the candidate answer tokens.

The main objective of the thesis is therefore to distinguish between three forms of multimodal behaviour: structural processing of audio tokens, measurable audio contribution to the output, and genuinely causal use of audio in the final decision. To address this objective, two independent but complementary analyses are used. The first applies the MM-SHAP framework, following Morais et al. [2], to estimate the relative contribution of audio and text. The second introduces audioDIME, an audio-domain adaptation of DIME that separates unimodal contributions from multimodal interactions.

2. Related Work

Multimodal Large Language Models (MLLMs). MLLMs extend language models by integrating non-textual modalities, such as images, audio, or video, through modality-specific encoders, alignment modules, and an

LLM backbone responsible for generation. Large Audio Language Models are a specific class of MLLMs in which audio is integrated with language, enabling tasks such as audio captioning, audio question answering, and Music Question Answering.

Explainable AI (XAI). To understand whether a response derives from the acoustic signal or from textual biases, observing the final output is not sufficient. In multimodal models, explainability must identify not only relevant features, but also which modality contributes to the prediction and whether modalities are actually integrated. This motivates the use of post-hoc methods based on perturbations and feature attributions.

MM-SHAP (Multimodal SHAP). MM-SHAP [3] extends SHAP to the multimodal context by estimating feature contributions and aggregating them by modality. It is performance-agnostic: it does not measure whether the answer is correct, but how much each modality contributes to the prediction. In this thesis, it is used to obtain a relative audio/text contribution score.

DIME (Disentangled Multimodal Explanations). DIME [1] is a local interpretability method for black-box multimodal models that separates Unimodal Contributions (UC) from Multimodal Interactions (MI). By decomposing the output into an additive unimodal component and a cross-modal residual, DIME makes it possible to distinguish modality dominance from genuine integration between modalities.

3. Methodology

Evaluation setting. The experiments are conducted on Qwen2.5-Omni-7B [5], focusing on the *Thinker* component, which performs multimodal understanding and textual generation. The model is evaluated on HumMusQA [4], a multiple-choice MQA benchmark composed of audio clips, questions, four candidate answers, and one correct answer. This format provides a controlled interpretative target, since the logits of the candidate answer tokens can be used as continuous scores for the explainability analyses.

General inference pipeline. Each sample is treated as an audio-text input: the audio is processed by the model encoder, while the question and options are tokenised into the multimodal

prompt. The candidate-answer logits provide the target score for two independent analyses: MM-SHAP, which estimates the relative contribution of audio and text, and audioDIME, which separates unimodal contributions from multimodal interactions.

3.1. MM-SHAP adaptation

The MM-SHAP adaptation follows the experimental logic of Morais et al. [2], redefining features and masking for the MQA setting. Textual features correspond to the question tokens, while scaffold and answer options are kept fixed to preserve the multiple-choice format. The waveform is divided into uniform temporal windows, with the number of audio segments matched to the number of perturbable question tokens. SHAP values are estimated through sampled permutations, aggregated by modality through absolute values, and normalised as:

$$\text{A-SHAP} = \frac{\Phi_A}{\Phi_A + \Phi_T}, \quad \text{T-SHAP} = 1 - \text{A-SHAP}. \quad (1)$$

This provides a compact measure of the relative dependence of the response on audio and text, without explicitly modelling interactions between modalities.

3.2. audioDIME adaptation

audioDIME adapts DIME to the generative audio-language setting considered in this thesis. Since Qwen2.5-Omni generates textual responses autoregressively, each target token is associated with a scalar score, defined as the pre-softmax logit assigned by the model:

$$m_k(x_a, x_t) = \ell_k[y_k]. \quad (2)$$

The score is then decomposed into a unimodal component and an interaction component:

$$m_k(x_a, x_t) = UC_k(x_a, x_t) + MI_k(x_a, x_t). \quad (3)$$

The two components are estimated through cross-combinations of audio-text pairs from a background set and then explained with LIME, separately by modality and contribution type, producing UC_{audio} , UC_{text} , MI_{audio} , and MI_{text} .

The audio representation is adapted to the musical domain using Demucs source separation and onset-based temporal segmentation. Each audio feature corresponds to a source-segment pair,

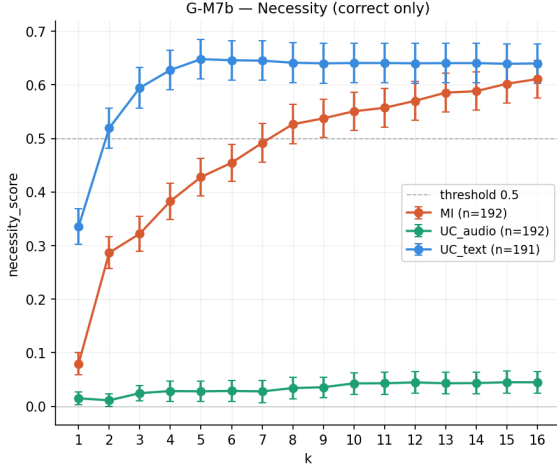
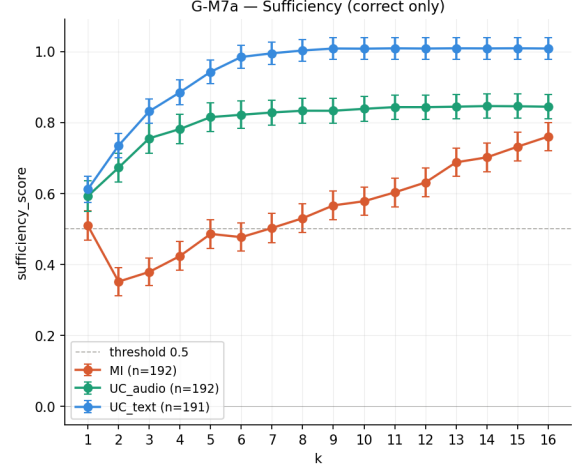
(a) Sufficiency for UC_{audio} , UC_{text} , and MI.(b) Necessity for UC_{audio} , UC_{text} , and MI.

Figure 1: Diagnostic comparison between unimodal contributions and multimodal interaction. Textual features are both sufficient and necessary, while unimodal audio features are largely replaceable; MI captures the regime in which audio becomes causally relevant through interaction with the question.

while textual perturbations are applied only to the question words, keeping scaffold and answer options fixed.

4. Experiments

Faithfulness of audioDIME explanations.

The first experiment evaluates whether audioDIME explanations are causally faithful: relevant features should preserve the response when retained and weaken it when removed. The analysis focuses on the top- k features ranked by Multimodal Interaction (MI), evaluated through chance-normalised sufficiency and necessity:

$$\text{suff.} = \frac{p_{\text{suf}} - p_{\text{chance}}}{p_{\text{orig}} - p_{\text{chance}}}, \quad \text{nec.} = \frac{p_{\text{orig}} - p_{\text{nec}}}{p_{\text{orig}} - p_{\text{chance}}}. \quad (4)$$

The threshold 0.5 indicates that at least half of the original informational advantage is preserved or destroyed.

In correct samples, MI explanations are more coherent: sufficiency reaches about 0.75 at $k = 16$, while necessity exceeds 0.5 around $k = 7$ and reaches about 0.61. In incorrect samples, the curves are flatter (suff. ≈ 0.43 – 0.52) and necessity exceeds the threshold only for larger subsets, suggesting decisions more often driven by few dominant or misleading features. The *MI balanced* diagnostic further shows that preserving a multimodal decision often requires both audio and textual features: the strongest MI features are not always the most sufficient subset.

The diagnosis in Fig. 1 clarifies the central result. UC_{text} is almost sufficient on its own and remains necessary, whereas UC_{audio} shows moderate sufficiency but almost null necessity. Unimodal audio can support the response, but is often compensated by text when removed. MI occupies an intermediate position: audio becomes causally relevant mainly when interpreted in relation to the question, not as an autonomous contribution. Overall, audioDIME provides locally causal explanations, while confirming that the causal role of audio is mainly interactional.

Aggregate modal behaviour. The second experiment aggregates MM-SHAP and audioDIME scores to characterise the modal behaviour of Qwen2.5-Omni on HumMusQA. For each sample, A -SHAP and T -SHAP are considered together with UC_a , UC_t , MI_a , and MI_t . The relative weight of audio-text interaction is summarised through:

$$\text{MI-ratio} = \frac{\|MI\|_1}{\|UC\|_1 + \|MI\|_1}, \quad \|v\|_1 = \sum_i |v_i|. \quad (5)$$

The L_1 norm aggregates attribution magnitudes, preventing positive and negative effects from cancelling out.

The results show a clear *text-first* pattern. In almost all conditions, T -SHAP is greater than A -SHAP, and the imbalance increases with difficulty: A -SHAP decreases from about 0.45 for low-difficulty questions to about 0.35 for high-difficulty ones, while the textual contribution

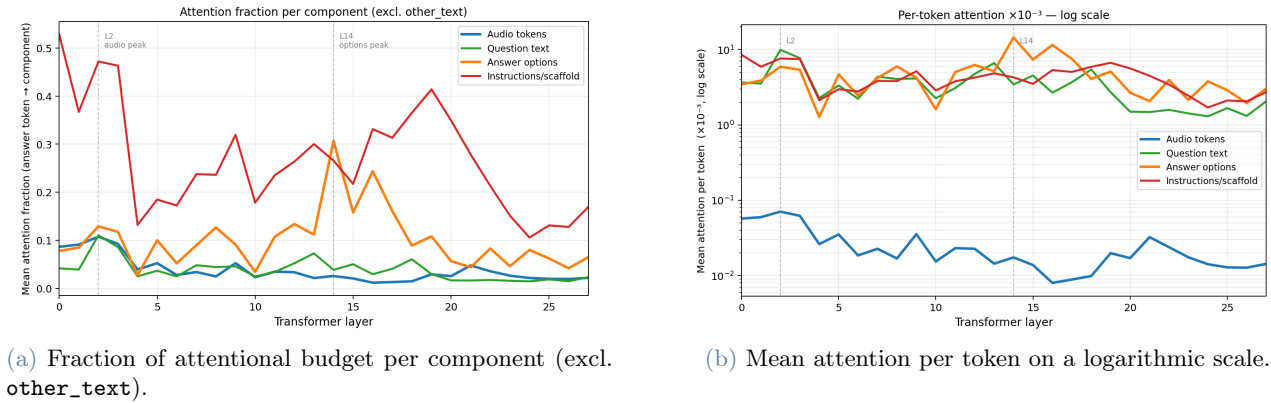


Figure 2: Layer-wise attention of the generated answer token to each input component.

risers to about 0.65. When acoustic information is harder to use, the model therefore relies more strongly on the question, candidate options, and linguistic priors.

audioDIME refines this result. UC_t remains high in both correct and incorrect samples, so text use alone is not an indicator of accuracy. The discriminative signal is instead multimodal interaction: correct responses have higher MI_a , MI_t , and MI-ratio, with centroids (0.43, 0.52) for correct samples and (0.38, 0.44) for incorrect ones. Questions in the dataset are divided into categories, and the results confirm the general trend, with Lyrics being the main exception (A -SHAP ≈ 0.60), suggesting stronger dependence on vocal audio but with high internal variability. Overall, text dominates the decision structure, while audio contributes more effectively when integrated with the question through multimodal interactions.

4.1. Inside the Network

The previous analyses show that Qwen2.5-Omni follows a predominantly text-first behaviour: text often supports the response, while audio becomes relevant especially through interaction with the question. To verify whether this pattern also emerges inside the model, the analysis moves to the *Thinker*, the transformer-decoder component responsible for multimodal understanding and textual generation.

The internal experiments are conducted on the 28 layers of the Thinker and on 960 HumMusQA evaluations, obtained from 320 samples through three answer-order permutations. Accuracy remains stable between original and per-

mutated configurations, from 66.56% to 66.46%, and 81.9% of samples produce the same response independently of option order. Audio tokens represent on average 93.1% of the input sequence, compared with 6.9% textual tokens, making it possible to distinguish between the structural presence of audio and its actual internal use.

Layers are denoted as L_X , with $X \in [0, 27]$.

Attention patterns. The first experiment measures, layer by layer, how much attention the decision token assigns to audio, instructions, question, options, and scaffold. As shown in Fig. 2a, a strong asymmetry emerges between the presence of audio and the attention it receives: although audio constitutes most of the input, audio tokens receive only 3.8% of the total attentional budget and, after per-token normalisation in Fig. 2b, remain about two orders of magnitude below textual components.

The layer-wise trend is dominated by text. Audio reaches its maximum at L2, with about 10.7% of attention, but then stabilises around 2.8%. Instructions are more relevant in early layers, options peak in the middle layers at about 30.7% at L14, and scaffold dominates the final layers. This progression suggests that the model moves from format and instruction constraints to a central comparison between textual alternatives, rather than to direct retrieval of acoustic information.

Correct samples show slightly higher audio attention in the middle and final regions, whereas incorrect samples assign more attention to the options in the peak window, suggesting possible answer picking. Since the curves remain almost

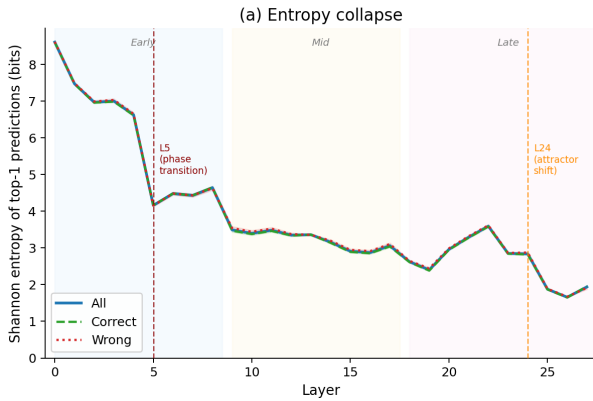


Figure 3: Analysis of audio-token representations across the layers.

overlapping across difficulty and category, the model does not increase audio attention when stronger acoustic grounding would be expected. The text-first bias observed post-hoc is therefore already visible in the internal attentional flows.

Logit Lens. The second experiment analyses the evolution of audio-token representations across the layers of the Thinker. At each layer, audio hidden states are projected through the language-model head, obtaining a distribution over the vocabulary. This analysis does not measure causal effects, but verifies whether audio representations remain lexically interpretable during the forward pass.

The main result is a progressive collapse towards non-semantic lexical attractors. Shannon entropy drops from 8.61 bits at L0, corresponding to about 383 effective tokens, to 1.66 bits at L26, fewer than 4 effective tokens. In parallel, Top-1 Dominance increases from 5.2% to 69.9%. The sharpest transition occurs between L4 and L5, followed by further concentration after L8. This phenomenon is systematic across correctness, difficulty, and category. It does not imply that audio information is absent, but shows that, once inside the Thinker, audio is no longer accessible as direct lexical content through the language-model head.

Metric	Early	Mid	Late
Top-1 (讓他們)	36.2%	59.2%	40.1%
Top-2 (Á)	8.6%	8.3%	34.0%
Cumulative Top-2	44.8%	67.5%	74.1%

Table 1: Concentration of the main attractors across the three depth regions of the model.

The collapse occurs mainly through two transitions. The first, between L4 and L5, abruptly reduces the number of effective tokens from approximately 100 to approximately 17, suggesting structural compression of the audio representations. The second, after L8, brings Top-1 Dominance above 50%, further concentrating the projections on a few attractors. The phenomenon is systematic: entropy curves are almost overlapping for correct and incorrect responses, and difficulty and category do not produce distinctive profiles. This result does not imply that audio information is absent, but shows that, once it enters the Thinker, it does not remain accessible as direct lexical content through the language-model head.

Attention knockout. The third internal experiment introduces a causal intervention on the attentional flows of the Thinker. For each layer window, the attention of the decision token towards a specific input component is blocked, and the resulting response is compared with the baseline. The analysed components are audio, question, options, prompt instruction, and a global `all_text_to_audio` block, which prevents textual tokens from attending to audio tokens.

The direct decision→audio channel is almost inert. Blocking audio changes the response in only 0.737% of cases and produces an accuracy variation of $\Delta acc = -0.141\%$, the lowest value among the analysed components. Moreover, audio knockout preserves 99.552% of already correct responses, indicating that most correct decisions remain unchanged even when the decision token cannot directly access audio.

The behaviour changes when the global `all_text_to_audio` pathway is blocked. The mean flip rate rises to 3.631%, with $\Delta acc = -0.729\%$, and the effect peaks at L14–L16, where the flip rate reaches 16.354% and accuracy drops by about 5.833 points. This window

coincides with the attention peak towards the options, suggesting that the model compares answer alternatives using audio as indirect context rather than as a source directly queried by the decision token.

Textual components instead become more relevant in the final layers: blocking the options peaks at 5.42% at L24–L26, while blocking the instruction reaches 6.35% in the same region, with $\Delta acc = -1.458\%$. Overall, attention knockout confirms the previous results causally: the final choice does not depend on strong direct access to audio, but on an indirect multi-modal configuration in the intermediate layers, followed by a final phase dominated by options and instructions.

5. Conclusions

This thesis shows that, on HumMusQA, a correct answer from Qwen2.5-Omni-7B does not necessarily imply genuine use of audio. The analyses converge towards a *text-first* behaviour: MM-SHAP highlights a textual contribution greater than the audio contribution, while audioDIME shows that textual features are sufficient and necessary, and that audio features are often replaceable.

The internal analyses confirm that audio is processed but rarely decisive. Although audio tokens constitute about 93% of the input, they receive only 3.8% of the attentional budget; Logit Lens shows a collapse towards non-semantic lexical attractors; attention knockout indicates that blocking direct access to audio has minimal effect, whereas interrupting the text-audio channel in the intermediate layers is more harmful. Overall, Qwen2.5-Omni uses audio mainly in a mediated way, through interaction with the question and the options, rather than as autonomous perceptual grounding. The main contribution of this work is therefore to distinguish between audio being present, audio being encoded, and audio being causally used: evaluating Large Audio Language Models requires analysing not only whether they answer correctly, but how the answer is produced.

6. Future Work

Future work should extend the analysis to open-ended music captioning, where free generation requires sequence-level or concept-level scoring

rather than multiple-choice logits. The same pipeline should also be applied to Audio LLMs with different encoders, fusion strategies, pre-training data, or musical specialisation, to test whether the weak causal role of audio is model-specific or general. Finally, audioDIME should be refined through alternative temporal granularities, source-separation strategies, and perceptually controlled masking.

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